1. Layer 1: $784 \times 256 = 200,704$ = MAC
   Layer 2: $256 \times 128 = 32,768$
   Layer 3: $128 \times 10 = 1280$

2. Sum of MAC = 234,752

3. Total number of trainable parameters = 234,752

4. Weight memory in bytes = $234,752 \times 4 = 939,008$

5. Total activation memory in bytes = $(4 \times 784) + (4 \times 256) + (4 \times 128) + (4 \times 10) = 4712$

6. Arithmetic intensity = $(2 \times 234,752) / (939,008 + 4712)$
   $= 0.4975$